Validation Section — Draft from the Full Validation Run (22 Oct 2026)

1 Validation

We evaluate AhToVIK with the three research questions introduced in the paper:

- **RQ1 (evidence-gated intervention).** Does evidence-gated intervention reduce unnecessary and repeated ineffective responses, and preserve mission continuity, compared with immediate-replacement, persistence-based and other related-work adaptation policies?
- **RQ2 (decision logic).** Does AhToVIK geometric compromise ranking form feasible teams efficiently and stably compared with alternative ranking and optimisation methods under the same eligibility constraints?
- **RQ3 (knowledge refinement).** Does reusing verified recovery cases shorten later recoveries under recurring comparable conditions compared with a frozen knowledge model, and what does it cost when the recorded history is misleading?

1.1 Experimental Setup

Mission model. All experiments use a synthetic search-and-rescue (SAR) mission model with a heterogeneous fleet of UAVs, UGVs and confined-space crawlers (20 %/60 %/20 % of the fleet in the dynamic benchmark). Robots are described by six criteria: navigation efficiency, payload, safety, communication, adaptability and energy (cost). Role weights come from AHP profiles with consistency ratio < 0.1. A task requires a complete team of six role slots (one scout, two rescuers, two carriers and one lifter). Experiments are implemented in Python 3.12 (NumPy 2.3, SciPy 1.17, HiGHS through `scipy.optimize.milp`). The master seed is 20261022. All configurations are *paired*: for a given seed, every method receives the same fleet, the same fault schedule and the same telemetry noise.

Dynamic world. Controllers never observe the ground truth. They see only noisy telemetry (relative Gaussian noise $\sigma = 0.05$), explicit fault flags and accepted progress checkpoints. A replacement costs three steps of handover. A quarter of the robots cannot resume another robot's checkpoint and must restart, losing half of the accumulated progress. Methods that use execution authority hold an 8-step lease with a 2-step clock-uncertainty margin. Diagnostics are conclusive with probability 0.95. A mission lasts 170 steps and its task deadline is step 115. Table 1 summarises the parameters.

Fault scenarios. The main dynamic benchmark covers ten ground-truth scenarios: *noise_only* (no faults); *transient* (2–3 short capability dips or telemetry dropouts); *degrade* (gradual capability ramp-down); *hard* (flagged robot failures); *silent* (robots stop reporting without a flag); *restriction* (a zone or mode restriction whose symptoms resemble a robot fault); *isolated* (a genuine robot fault with the same symptoms); *stale_competing* (a silent failure of the top standby robot followed by two hard failures); *cascade* (three successive failures on the same slot); and *mixed* (three distinct fault types combined). The resilience sweeps vary one stress factor at a time on the *mixed* scenario: 0–6 injected faults, telemetry noise $\sigma \in \{0.02, 0.05, 0.1, 0.2\}$, packet loss $\in \{0, 0.05, 0.15\}$ and fleet size $\in \{14, 18, 22, 28\}$ robots.

Table 1: Experimental parameters

Parameter	Value
Static benchmark	
Fleet size (core / scarcity)	100 / 200 robots
Supply ratios (scarcity)	$0.8\times, 1\times, 1.5\times, 3\times, 10\times$
Scalability grid	$N = 10 \ldots 10{,}000$ (500 to 15 seeds per N)
Runs per case	500 paired instances
Dynamic benchmark	
Fleet / team	28 robots / 6 role slots
Horizon / deadline	170 / 115 steps
Telemetry noise / handover	$\sigma = 0.05$ / 3 steps
Lease / clock margin	8 / 2 steps
Diagnostic accuracy	0.95
Scenarios × methods × runs	$10 \times 12 \times 500 = 60{,}000$
Sweep runs	93,500
Knowledge-loop benchmark	
Streams × sequences × episodes	$4 \times 500 \times 8$

Knowledge-loop streams. Each sequence uses a fixed fleet and eight consecutive missions (episodes). Every episode contains one interruption that can be resolved only once its cause is known. A zone/mode restriction is resolved by reconfiguring the incumbent robot, and an isolated fault by replacing it. The four streams control how informative the recorded history is. In *recurring*, the restriction occurs in 90 % of episodes, so history should help. In *drift*, the restriction occurs in 90 % of the first half of the stream and never afterwards, so the prior becomes wrong. In *spurious*, the first two episodes are restrictions and later ones only 20 %, so early cases mislead. In *nonrecurring*, slot, zone and cause are random, so contexts rarely match.

1.2 Baselines

Table 2 lists the compared methods. The dynamic baselines are compact re-implementations of each related-work mechanism. They run on the identical plant, telemetry, checkpoint and handover substrate as AhToVIK, and none of them has an evidence gate, a diagnostic layer, expiring authority or a bounded interruption budget unless stated. The static baselines are the allo-

cation cores of the dynamic ones. CO-MILP corresponds to the global assignment used by Mayya, Calvo and Filippone, and greedy or auction allocation corresponds to the initial allocation of the remaining methods. Two AhToVIK ablations isolate our contributions. *AhToVIK-nogate* removes the evidence gate and diagnostics. *AhToVIK-frozen* disables knowledge reuse.

1.3 Metrics and Statistical Tests

Mission continuity. *On-time rate*: share of tasks completed before the deadline, averaged over missions (not the share of fully successful missions). *Worst-case on-time*: lowest per-scenario mean. *Progress at deadline*: share of task work completed by the deadline. *Downtime*: slot-steps without progress.

Intervention quality (RQ1). *Unnecessary reassignments*: replacements of robots that were not actually faulty. *Repeated ineffective interventions*: further interventions on a slot after an earlier one failed to restore progress. *Work lost*: progress discarded by restarts. *Dual-authority events*: two robots executing the same slot at once. *Duplicate dispatches*.

Decision quality (RQ2). Latency (build and solve), log–log scalability slope, objective gap to CO-MILP, slot fill rate and complete-feasible-team rate under scarcity, priority-weighted unmet demand, rank stability under the VIKOR weight v (Kendall τ, team Jaccard), and determinism under robot-order permutation.

Knowledge reuse (RQ3). Recovery latency (steps from interruption to restored progress), diagnostic steps, downtime and on-time rate, averaged over episodes 3–8 (the first two episodes are excluded as warm-up).

Tests. Runs are paired by seed, so we use the two-sided Wilcoxon signed-rank test and apply the Holm–Bonferroni correction within each table ($\alpha = 0.05$). We report the Vargha–Delaney $\hat{A}_{12}$ effect size, the probability that AhToVIK yields the *higher* value. For lower-is-better metrics, $\hat{A}_{12} < 0.5$ therefore favours AhToVIK. We read $|\hat{A}_{12} - 0.5| \geq 0.21$ as a large effect. Static results include bootstrap 95 % confidence intervals. Pooled over all ten scenarios, many runs contain zero interventions for every method, which pulls pooled $\hat{A}_{12}$ towards 0.5. We therefore also report per-scenario effects where the mechanisms differ.

1.4 Results

1.4.1 RQ1 — Evidence-gated intervention and mission continuity

Fewer and better-targeted interventions. The evidence gate nearly eliminates ineffective adaptation (Fig. 1, Table 3). AhToVIK performed 0.027 repeated ineffective interventions per mission. This is 86 % fewer than the best baseline, RAINBOW (0.187), and 98–99 % fewer than every other method (1.55–3.16). It also had the fewest unnecessary reassignments (0.074 per mission, half of RAINBOW and RODRIGUES and 6 % of FILIPPONE), and it lost the least work to restarts (0.96 work units against 5.89–28.43, an 84 % reduction relative to the best baseline). All of these differences are significant against every baseline ($p_{\text{Holm}} < 10^{-4}$).

The ablation isolates the gate as the cause. With the same decision logic, AhToVIK-nogate makes 66 times more repeated ineffective interventions (1.78), loses 8.8 times more work and completes 5.3 percentage points (pp) fewer tasks on time (85.4 % vs. 90.7 %, $p_{\text{Holm}} < 10^{-4}$).

The gain is concentrated where fault symptoms are ambiguous. In the *restriction* scenario, a zone/mode restriction looks like a robot fault. Baselines replace the incumbent, the replacement meets the same restriction, and the cycle repeats: 9.2–20.6 repeated interventions per mission for every method except RAINBOW (1.17). AhToVIK diagnoses the cause first and reconfigures the incumbent, reducing this to 0.02 ($\hat{A}_{12} = 0.05$ vs. MARKET, a large effect). In this scenario it also halves downtime relative to MARKET (45.9 vs. 89.3 steps, $\hat{A}_{12} = 0.30$) and achieves the highest on-time rate of all methods (80.6 % vs. 79.3 % for RAINBOW and $\leq$70.7 % for the others). In the *transient* scenario, the gate waits out self-clearing dips. Unnecessary reassignments fall from 1.94 (MARKET) and 1.13 (RAINBOW) to 0.54, and downtime from 18.6 and 12.6 to 5.9 steps ($\hat{A}_{12} = 0.12$ and 0.27).

Safe handover. Authority leases and atomic reservation produced zero duplicate dispatches across all 29,500 AhToVIK simulations (main, sweeps and knowledge streams), compared with 0.18–1.81 per mission for the six baselines that lack them. Dual-authority events fell from 0.135–0.185 per mission to one event in 5,000 main-benchmark missions (see Section 1.5).

Robustness under compound faults and noisy telemetry. AhToVIK has the best *worst-case* on-time rate (77.1 % vs. 75.7 % for RAINBOW and $\leq$70.7 % for all others). It leads in the three hardest scenarios (Fig. 2): *restriction* (+1.3 to +42.3 pp), *cascade* (+2.7 to +24.3 pp) and *mixed* (+1.3 to +31.1 pp). Its progress at the deadline (98.3 %) is significantly higher than that of every baseline except RAINBOW. The sweeps (Fig. 3) show that this advantage grows with stress. From four injected faults upwards, AhToVIK has the highest on-time rate (60.1 % vs. 58.9 % for RAINBOW and 55.7 % for MARKET at six faults). It is also the least sensitive method to telemetry noise. Between $\sigma = 0.02$ and 0.2 it loses only 0.9 pp (76.8→75.9 %), whereas the immediate-replacement MARKET collapses from 71.7 % to 50.5 % because noise-induced false alarms trigger replacements. With 15 % packet loss, AhToVIK stays at 76.9 % while MARKET drops to 64.9 %. Overall, AhToVIK ranks second on the resilience score (area under the sweep curves: 0.926, behind RAINBOW at 0.938). Its decision overhead is the lowest per decision (0.32 ms, p95 5.2 ms).

1.4.2 RQ2 — Decision logic

Feasibility-first team formation. Under scarcity, AhToVIK forms the most complete feasible teams (Fig. 4b, Table 4). At $0.8\times$ supply it fully staffs 51.5 % of tasks, against 40.3 % for CO-MILP and about 46.3 % for every greedy method, a relative gain of 28 % over the optimiser. The reason is structural. The objective of CO-MILP and of the greedy methods rewards filling *slots*, so they spread scarce robots across many partially staffed tasks that cannot start. AhToVIK commits robots only to teams it can complete.

Scalability. Decision time grows sub-linearly with fleet size

Table 2: Compared methods. Dynamic baselines are mechanism-inspired re-implementations, not the original systems.

Benchmark	Method	Mechanism
Static (RQ2)	**AhToVIK**	Geometric reference mapping, compromise index and bounded depth-first formation of complete teams.
	AhToVIK-core / -partial	AhToVIK ranking without the eligibility gate / with partial staffing allowed.
	CO-MILP	Exact combinatorial optimisation (MILP, HiGHS) with a BIG_M fill-first term. Certified objective ceiling.
	SAW / TOPSIS / VIKOR-greedy	Classical MCDM score, then greedy assignment [15][16].
	TRAIT-greedy	Trait-coverage matching [5][14].
	SSI-auction	Sequential single-item auction.
	RANDOM	Uniform random feasible assignment (lower bound).
Dynamic (RQ1, RQ3)	**AhToVIK**	Interruption deadline, ordered adaptation policy, evidence gate, authority leases, atomic reservation, verification and knowledge reuse.
	AhToVIK-nogate	Same decision logic, no evidence gate or diagnostics (acts on persistent deviation).
	AhToVIK-frozen	Same as AhToVIK but recorded cases are never reused.
	MARKET	Immediate replacement through a sequential auction on visible utility [13].
	RAINBOW	Architecture-based adaptation: design-time ordered repair strategy with retries (persistence-based trigger) [7][8][17][19].
	MAYYA	Capability estimation from observed performance and global optimal reallocation [5].
	CALVO	Online local repair with periodic global recomputation and hysteresis [6].
	FILIPPONE	Trigger functions with a backup coalition per task [12].
	ESFAHANI	Uncertainty-aware: acts on the posterior probability of failure [3].
	CARWEHL	Uncertainty reduction: adds an information-acquisition probe when the posterior is ambiguous [10].
	CAMARA	Latency-aware proactive adaptation from the capability trend [11].
	RODRIGUES	Learning-based refinement: learns cause statistics online and acts on them [20].

Table 3: Main dynamic benchmark: macro-average over ten scenarios × 500 paired seeds (sorted by on-time rate). Best per column in bold. AhToVIK-frozen is identical to AhToVIK here because every single mission starts with an empty knowledge base.

Method	On-time (%)↑	Worst-case on-time (%)↑	Progress at deadline (%)↑	Downtime (steps)↓	Unnecessary reassign.↓	Repeated ineffective↓	Work lost ↓	Dual-authority↓	Duplicate dispatch↓
RAINBOW	**92.4**	75.7	**98.7**	**21.8**	0.150	0.187	5.89	0.138	0.179
ESFAHANI	91.6	70.4	97.9	23.9	0.195	2.759	8.13	0.138	1.356
CARWEHL	91.6	70.3	98.0	24.0	0.203	2.444	7.90	0.138	1.661
MARKET	91.3	70.3	98.0	24.5	0.292	2.571	8.24	0.185	1.806
RODRIGUES	91.0	70.0	98.0	25.1	0.152	1.664	7.82	0.139	1.040
CAMARA	90.9	70.2	97.6	25.3	0.383	3.164	9.60	0.154	0.227
FILIPPONE	90.7	70.7	97.7	26.9	1.277	3.141	12.38	0.169	**0.000**
AhToVIK	90.7	**77.1**	98.3	25.7	**0.074**	**0.027**	**0.96**	**0.0002**	**0.000**
AhToVIK-nogate	85.4	65.2	97.0	26.6	0.136	1.781	8.47	0.135	**0.000**
CALVO	82.1	61.9	96.0	38.6	2.557	2.989	22.04	0.143	**0.000**
MAYYA	79.6	38.3	95.5	55.7	7.468	1.552	28.43	0.137	**0.000**

Pooled paired Wilcoxon (Holm): AhToVIK is significantly lower than every baseline on unnecessary reassignments, repeated ineffective interventions, work lost and dual authority ($p_{\text{Holm}} < 10^{-4}$).

(fitted log–log slope 0.43, against 1.04 for CO-MILP; Fig. 4a). At $N = 10{,}000$ robots, AhToVIK decides in 13.4 ms, while CO-MILP needs 8.7 s (about 650× slower). AhToVIK is also faster than TOPSIS- and VIKOR-greedy at that size (21.9 and 22.4 ms). At $N = 100$, its 1.5 ms is dominated by building the ranking (1.2 ms), whereas CO-MILP spends 32.7 of its 34.3 ms in the solver.

Stability and determinism. The geometric compromise ranking is more stable under changes of the VIKOR weight v than classical VIKOR (Kendall τ 0.89 vs. 0.82; team Jaccard 0.88 vs. 0.80). It is fully deterministic, since permuting the robot order never changes its solution. CO-MILP, by contrast, returns a different but equally optimal solution in 75.8 % of instances when robots are permuted, and in 69 % when the objective is nudged by 10^{-9}. This solution multiplicity makes optimiser-based allocation hard to audit and reproduce.

1.4.3 RQ3 — Knowledge refinement

Learning from recurring conditions. When a condition recurs, reusing verified cases pays off quickly (Fig. 5, Table 5). In the *recurring* stream, recovery latency falls from 19.6 steps in episode 1 to 14.5 by episode 8. Over episodes 3–8 it is 25.4 % lower than with frozen knowledge (14.9 vs. 20.0 steps, $\hat{A}_{12} = 0.15$, a large effect). The mechanism is visible in the diagnostic effort: ordering diagnostics by recorded cause frequency cuts diagnostic steps from 15.8 to 11.2 and downtime from 39.7 to 29.2 steps. For interruptions actually caused by a restriction, recovery takes 14.8 instead of 20.8 steps. The knowledge base grows to 13.9 verified cases per sequence by episode 8. On *nonrecurring* streams, reuse causes no regression (16.5 vs. 16.7 steps, a 1.5 % improvement). Compared with methods that adapt without verified diagnosis (no-gate, RODRIGUES, MARKET), AhToVIK recovers 2.4–4.8 times faster in the *recurring* and *nonrecurring* streams. This includes RODRIGUES, which learns cause statistics online but acts on

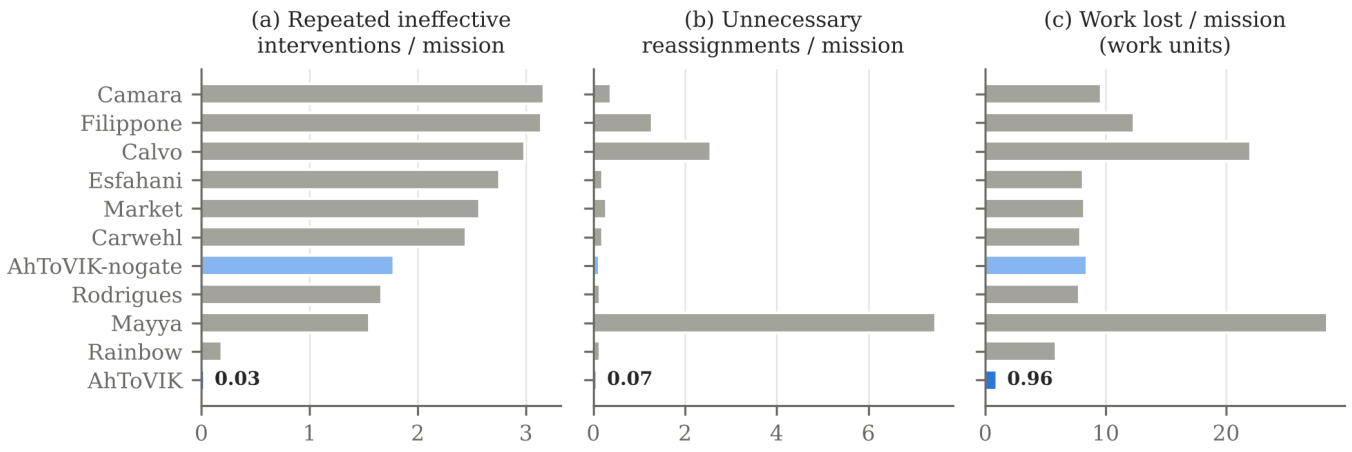


Figure 1: Intervention quality over all 5,000 missions per method (10 scenarios × 500 seeds). Dark blue: AhToVIK; light blue: the no-gate ablation; grey: related-work baselines.

Table 4: Static decision benchmark (500 paired instances; $N = 100$ unless noted). Gap: objective gap to CO-MILP. Teams/Slots: complete feasible teams / feasible slots filled at $0.8\times$ supply. τ: rank stability over v. Perm.: instances whose solution changes under robot-order permutation.

Method	ms	Slope	Gap (%)↓	Teams (%)↑	Slots (%)↑	τ ↑	Perm. (%)↓
AhToVIK	1.53	**0.43**	7.42	**51.5**	65.1	**0.89**	**0.0**
CO-MILP	34.28	1.04	**0.00**	40.3	**68.2**	—	75.8
SAW-greedy	1.03	0.51	4.14	46.3	**68.2**	—	**0.0**
TOPSIS-greedy	5.13	0.57	5.91	46.3	**68.2**	—	**0.0**
VIKOR-greedy	1.35	0.55	4.57	46.3	**68.2**	0.82	**0.0**
TRAIT-greedy	0.71	0.46	6.66	46.4	**68.2**	—	**0.0**
SSI-auction	**0.12**	0.56	0.38	46.4	**68.2**	—	**0.0**
RANDOM	0.31	—	23.38	—	—	—	—

AhToVIK gap 95 % CI: 7.05–7.80 %. All 500 CO-MILP runs returned an optimal status.

Table 5: Knowledge-loop benchmark: mean recovery latency (steps) over episodes 3–8, 500 sequences per stream. Δ: AhToVIK vs. frozen (negative = faster). $\hat{A}_{12}$: AhToVIK vs. frozen; all $p_{\text{Holm}} \leq 0.0013$.

Stream	AhToVIK	Frozen	Δ (%)	$\hat{A}_{12}$	No-gate	RODRIGUES
recurring	**14.9**	20.0	**−25.4**	0.15	64.8	63.0
nonrecurring	**16.5**	16.7	−1.5	0.47	39.2	39.2
drift	15.5	**15.0**	+3.2	0.69	26.0	24.3
spurious	15.6	**14.4**	+8.5	0.74	19.7	17.9

them without verification (63.0 steps in *recurring*, against 14.9 for AhToVIK).

1.5 Negative Results and Their Causes

The evaluation also exposes clear costs. We report them because they mark the boundaries of the approach, and because most of them follow directly from deliberate design choices.

(N1) Average on-time rate is not the highest. Macro-averaged over all scenarios, AhToVIK's on-time rate (90.7 %) is 1.7 pp below RAINBOW's and significantly below MARKET, ESFAHANI and CARWEHL. The effects are small ($\hat{A}_{12} = 0.48$–0.49) but consistent (Table 3). RAINBOW also has lower average downtime (21.8 vs. 25.7 steps). The loss comes from the simple scenarios in the upper rows of Fig. 2, explained in N2–N4, and it is outweighed only in the ambiguous and compound scenarios. AhToVIK therefore trades a few points of average-case performance for a better worst case and far fewer disruptive interventions. It is not a uniform improvement.

Table 6: Scenarios where AhToVIK is worse (on-time %, downtime in steps). Downtime differences vs. MARKET and RAINBOW are significant with Holm correction, except in *noise_only*.

	On-time (%)			Downtime (steps)		
Scenario	AhToVIK	MARKET	RAINBOW	AhToVIK	MARKET	RAINBOW
noise_only	95.4	99.7	99.5	2.2	0.5	0.5
isolated	94.3	99.3	99.4	11.2	3.6	7.0
silent	91.9	94.9	95.3	33.9	15.4	18.5
stale_competing	89.5	94.6	95.3	30.1	13.3	16.4
degrade	94.7	97.5	96.7	15.3	8.8	10.6

(N2) Fault-free missions are slower. With no faults (*noise_only*), AhToVIK completes 95.4 % of tasks on time, against 99.5–99.7 % for the baselines. Its mean task finish time (92.6 steps, against 89.0 for RAINBOW) is higher than that of every baseline except CALVO and MAYYA. The compromise index selects *balanced* robots that score acceptably on all six criteria, including safety and energy, rather than the fastest robot that the utility-greedy initial allocation of the baselines picks. The compromise ranking is therefore not utility-optimal when nothing goes wrong. A utility-weighted tie-break among top-ranked candidates would recover part of this gap.

(N3) Waiting costs for clear-cut faults. When a failure is unambiguous, the safeguards add delay without benefit (Table 6). In *silent* and *stale_competing*, AhToVIK must wait for the failed robot's authority lease to expire (8 steps plus a 2-step clock margin) before reassigning its slot, so that two robots never execute the same slot. Downtime is therefore about twice that of MARKET (33.9 vs. 15.4 and 30.1 vs. 13.3 steps;

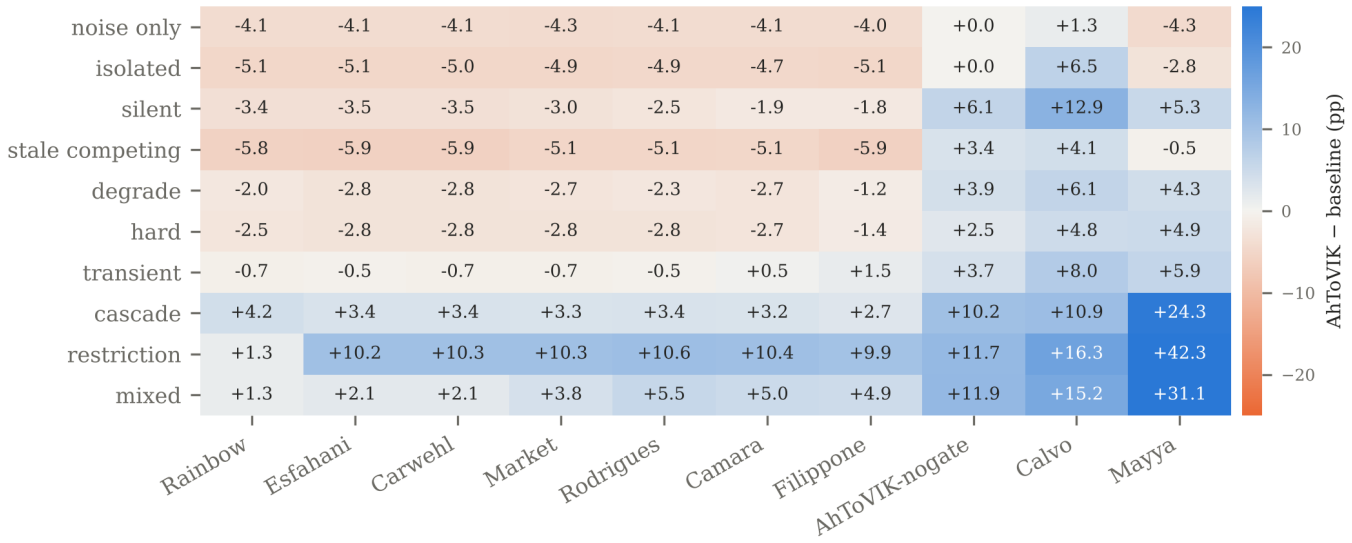


Figure 2: On-time rate difference (percentage points) between AhToVIK and each baseline per scenario. Blue: AhToVIK better; orange: AhToVIK worse. Rows are ordered from fault-free and single-fault scenarios (top) to ambiguous and compound scenarios (bottom).

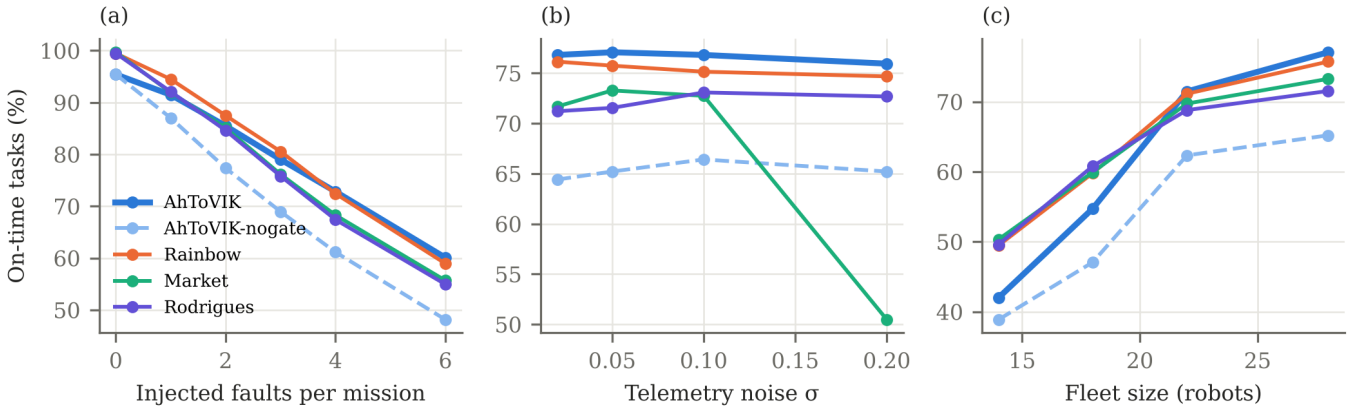


Figure 3: Resilience sweeps on the *mixed* scenario (500 paired seeds per point). (a) Increasing fault count, (b) telemetry noise, (c) fleet size.

$\hat{A}_{12} = 0.79$ in *silent*). In *isolated* and *degrade*, the diagnostic needed to rule out a restriction delays a replacement that immediate-replacement methods make straight away (downtime 11.2 vs. 3.6 steps, $\hat{A}_{12} = 0.98$). This is the price of avoiding dual authority and repeated ineffective responses in the scenarios where symptoms mislead. Shorter leases or an immediate path for explicit hardware fault flags would reduce it.

(N4) Small fleets. With 14 or 18 robots, AhToVIK is the worst method except its ablation (Fig. 3c). Its on-time rate is 42.0 % and 54.7 %, against 49.5–50.3 % and 59.8–60.8 % for RAINBOW, MARKET and RODRIGUES. It overtakes them only from 22 robots upwards. Because AhToVIK staffs complete teams only, a lack of spare robots leaves tasks waiting for a full team, whereas the baselines start tasks partially staffed and make some progress. The partial-staffing variant (AhToVIK-partial) is a candidate remedy for very small fleets, but it was evaluated only in the static benchmark.

(N5) Larger objective gap than simpler heuristics. AhToVIK's objective gap to CO-MILP (7.42 %, 95 % CI 7.05–7.80) is larger than that of SAW-, VIKOR- and TOPSIS-greedy (4.1–5.9 %) and of SSI-auction (0.4 %). Under scarcity, its slot fill rate (65.1 %) is also below the 68.2 % of all other methods, and its priority-weighted unmet demand is higher (0.267 vs. 0.228). Both effects come from the complete-team rule. Robots that could fill isolated slots of tasks that cannot be completed are left unassigned, which lowers slot-based metrics while raising the number of tasks that can actually start (51.5 % vs. 40.3 %). Which method is better therefore depends on the metric. CO-MILP's objective measures slot utility, not executable teams, so it is not a certificate that both methods solve the same constrained problem.

(N6) Misleading history costs a little. When the recorded prior is wrong, reuse is slower than frozen knowledge: +3.2 % in *drift* ($\hat{A}_{12} = 0.69$) and +8.5 % in *spurious* ($\hat{A}_{12} = 0.74$).

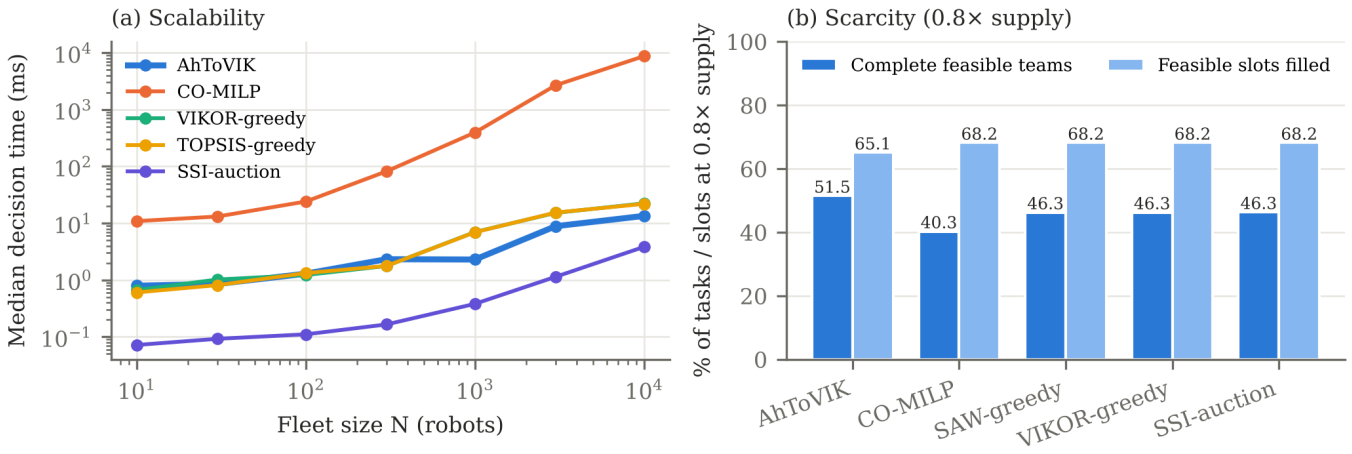


Figure 4: Static decision benchmark. (a) Median decision time vs. fleet size (log–log). (b) Complete feasible teams and feasible slots filled at $0.8\times$ supply (500 instances, 200 robots).

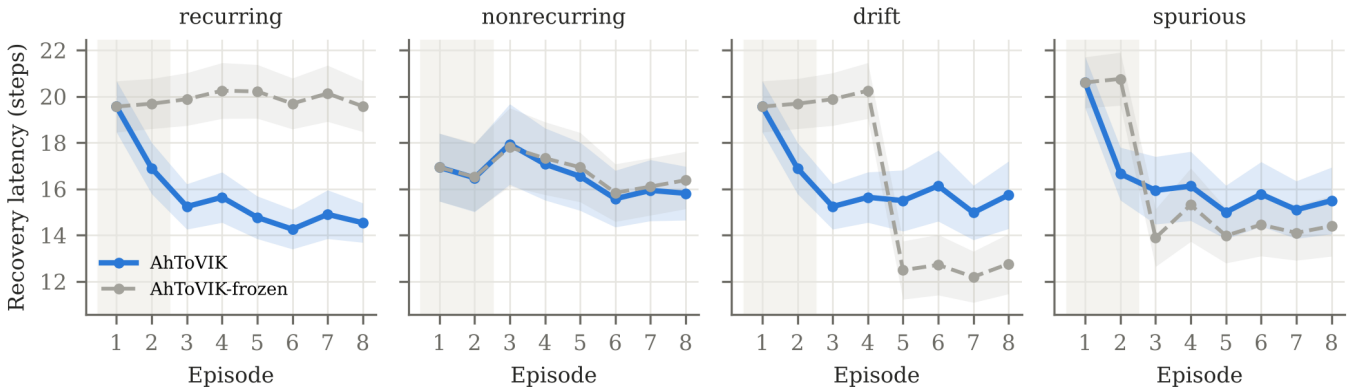


Figure 5: Recovery latency per episode for AhToVIK with knowledge reuse vs. the frozen ablation (500 sequences per stream; bands: 95 % CI). The shaded episodes 1–2 are warm-up and are excluded from Table 5.

Fig. 5 shows the cause. After the drift point (episode 5), every interruption is an isolated fault. The frozen model, whose default diagnostic order tests this cause first, drops to about 12.5 steps, while AhToVIK keeps testing the historically frequent restriction first and stays at about 15.5 steps. The knowledge base accumulates cause frequencies without forgetting, so stale evidence keeps influencing the diagnostic order. The same effect appears per cause in the *recurring* stream. The rare isolated faults take 15.5 steps with reuse against 12.9 without. The harm is bounded, since AhToVIK never becomes slower than its first-episode latency and remains 1.15–1.7 times faster than the no-gate ablation and RODRIGUES in these streams. A recency-weighted or windowed prior is a natural fix.

(N7) One dual-authority event. A zero-conflict claim is *not* supported. One reproducible dual-authority event occurred in the main benchmark (scenario *transient*, seed 128), and 16 occurred across 8,500 sweep simulations: 10 at telemetry noise $\sigma \geq 0.1$, 5 with two or more injected faults and 1 with 14 robots. No events occurred in the 16,000 knowledge-stream simulations. The rate (0.0002 per mission in the main benchmark) is about 700 times lower than that of the baselines, but non-zero. In 16 of the 17 affected runs, the event coincides with an unnecessary reassignment. This suggests that a noise-induced false alarm passed the evidence gate while the previous executor still held a valid lease.

(N8) Implementation cost and escalations. The AhToVIK runtime has 235 lines of core logic and 12 tunable parameters, against 43–73 lines and 3–8 parameters for the baselines. It also escalates to a human operator 0.49 times per mission when the bounded interruption budget is exhausted. Baselines never escalate. They keep retrying automatically, which accounts for part of their repeated ineffective interventions. Escalation is intended behaviour, but it places a supervisory load on the operator that a deployment must account for.

1.6 Threats to Validity

Construct. The related-work baselines are compact, mechanism-inspired re-implementations on a shared substrate, not reproductions of the original systems, and their absolute numbers should not be attributed to those systems. The recorded

"makespan" field is the mean task finish time (with the horizon for unfinished tasks), not the maximum. **Internal.** All methods are paired by seed and use the same fleet, faults and noise. Two execution fixes were applied to the notebook: loky-based parallel workers, and a check that FILIPPONE's backup team contains every active role. Algorithms, seeds and sample sizes were otherwise unchanged, and all 15 sanity checks passed. **Conclusion.** With 500 paired runs, even small differences reach significance, so we base claims on effect sizes and per-scenario trade-offs rather than on p-values alone. **External.** Results come from a synthetic SAR simulation with a closed-world two-cause diagnostic model. Wall-clock timings were collected in a shared container and are environment-dependent. The scalability sweep increases fleet size with a fixed number of task templates, so it does not establish scaling under jointly increasing demand.